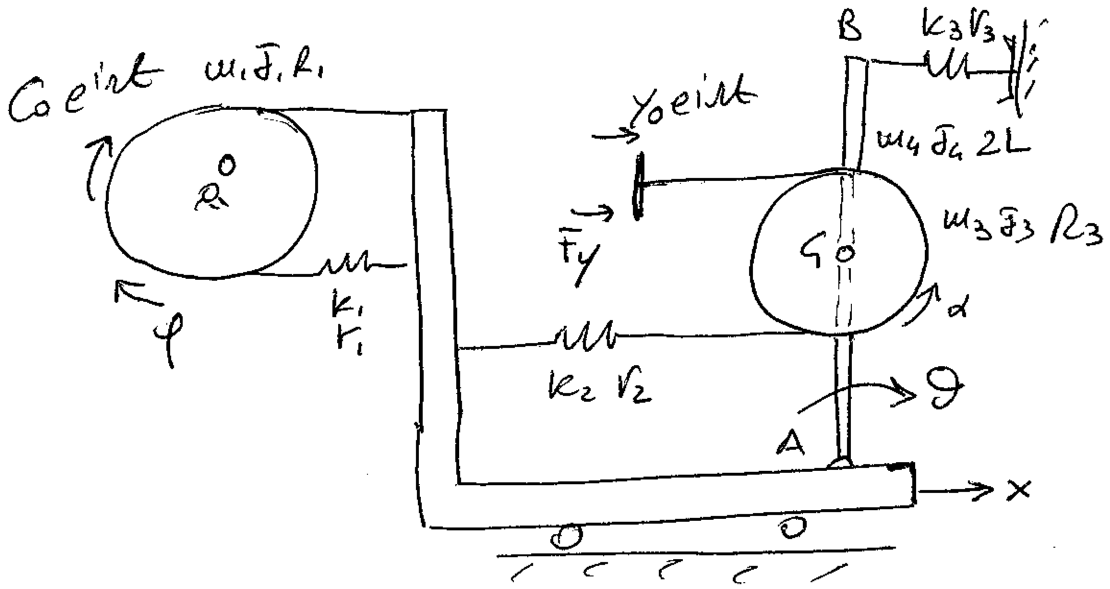
APPELLO 5 dic 13 - A -

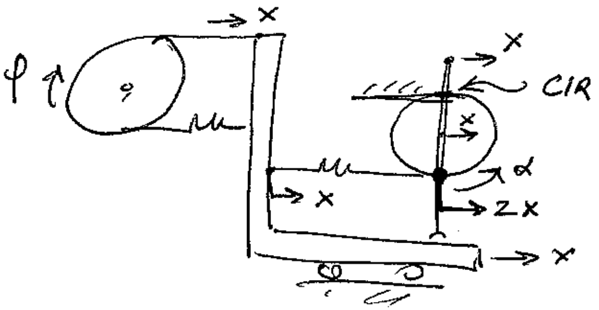


$$x = \begin{Bmatrix} x \\ \theta \\ Y \end{Bmatrix}$$

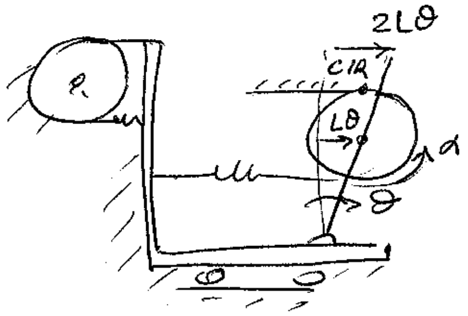
$$E_c = \frac{1}{2} J_1 \dot{\varphi}^2 + \frac{1}{2} m_2 \dot{x}^2 + \frac{1}{2} (m_3 + m_4) \dot{x}_G^2 + \frac{1}{2} J_4 \dot{\theta}^2 + \frac{1}{2} J_3 \dot{\alpha}^2$$

$$V_{EL} = \frac{1}{2} k_1 \Delta l_1^2 + \frac{1}{2} k_2 \Delta l_2^2 + \frac{1}{2} k_3 \Delta l_3^2$$

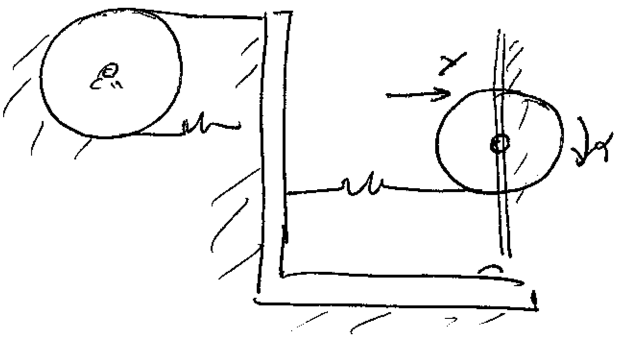
$x \neq 0 ;\ \vartheta = Y = 0$



$x = Y = 0 ;\ \vartheta \neq 0$



$x = \vartheta = 0 ;\ Y \neq 0$



$$\alpha = \frac{x}{R_1}$$

$$x_G = x + L\theta$$

$$\alpha = \frac{x}{R_3} + \frac{L\theta}{R_3} - \frac{Y}{R_3}$$

$$\Delta l_1 = 2x$$

$$\Delta l_2 = 2L\theta + x - Y$$

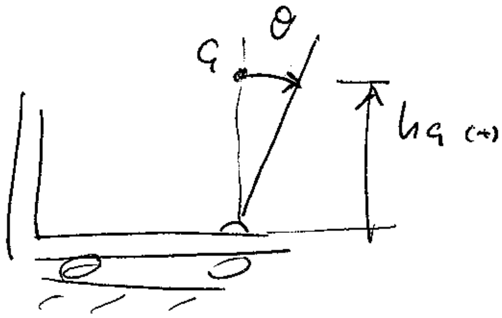
$$\Delta l_3 = -x - 2L\theta$$

$$m_F = \mathrm{diag}\left(J_1 \quad m_2 \quad m_3+m_4 \quad J_4 \quad J_3 \right)$$

$$k_F = \mathrm{diag}\left(k_1 \; k_2 \; k_3 \right)$$

$$r_F = \mathrm{diag}\left(r_1 \; r_2 \; r_3 \right)$$

$$\Lambda_m = \begin{bmatrix} 1/R_1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & L & 0 \\ 0 & 1 & 0 \\ 1/R_3 & L/R_3 & -1/R_3 \end{bmatrix} \qquad \Lambda_k = \Lambda_r = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2L & -1 \\ -1 & -2L & 0 \end{bmatrix}$$



$$V_G = (m_3+m_4)\, g\, h_G$$

$$h_G = L\cos\vartheta$$

$$\frac{\partial^2 V_G}{\partial \theta^2} = -(m_3+m_4)\, g\, L\cos\vartheta$$

$$[k_G] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -(m_3+m_4)gL & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

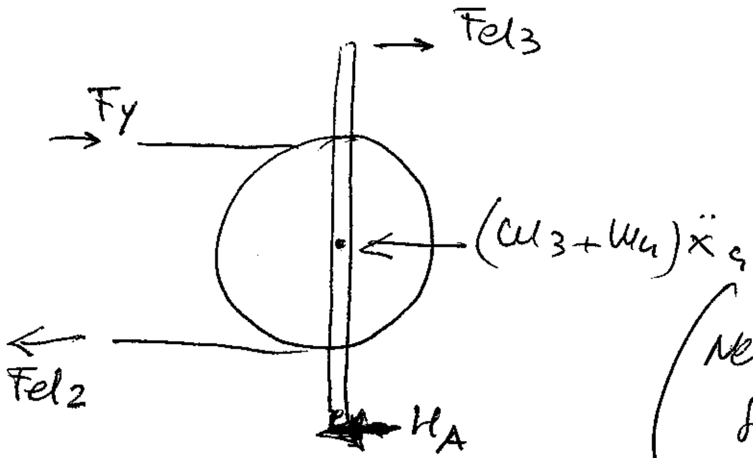
$$\delta L = C\,\delta\varphi + F_y\,\delta y = \frac{C}{R_1}\,\delta x + F_y\,\delta y$$

$$\rightarrow \quad Q_C = \begin{Bmatrix} 1/R_1 \\ 0 \\ 0 \end{Bmatrix} C e^{i\Omega t} \quad ; \quad Q_y = \begin{Bmatrix} 0 \\ 0 \\ F_y \end{Bmatrix}$$

$$F_{el_3} = (k_3 + i\Omega r_3)\,\Delta l_3 \qquad \text{(domanda 4)}$$

$$F_y = Q_V = [M_{VL}]\ddot{x}_L + [R_{VL}]\dot{x}_L + [k_{VL}]x_L + [M_{VV}]\ddot{x}_V + [R_{VV}]\dot{x}_V + [k_{VV}]x_V$$

(domanda 5)



(Nel disegno sono indicate solo le forze orizzontali già linearizzate)

$$Fel_3 = (k_3 + i\Omega r_3)\,\Delta L_3$$
$$Fel_2 = (k_2 + i\Omega r_2)\,\Delta L_2$$

(positive per $\Delta L_2 > 0$ $\Delta L_3 > 0$ → trazione sulle molle → forze sulla struttura con il principio di azione e reazione)

$$\ddot{x}_G = -\Omega^2 x_G$$

$$\Sigma F_o = 0 \; : \quad Fel_3 + F_y - Fel_2 - (m_3+m_4)\ddot{x}_G - H_A = 0$$

$$\boxed{H_A = Fel_3 + F_y - Fel_2 - (m_3+m_4)\ddot{x}_G}$$

NB: F_y si calcola come al punto precedente, ma non è lo stesso del punto precedente perché è applicata anche la coppia Coriolis